Three Ensembles, One Exponent

Microcanonical, canonical, and grand canonical thermodynamics of an ideal Fermi gas in d dimensions

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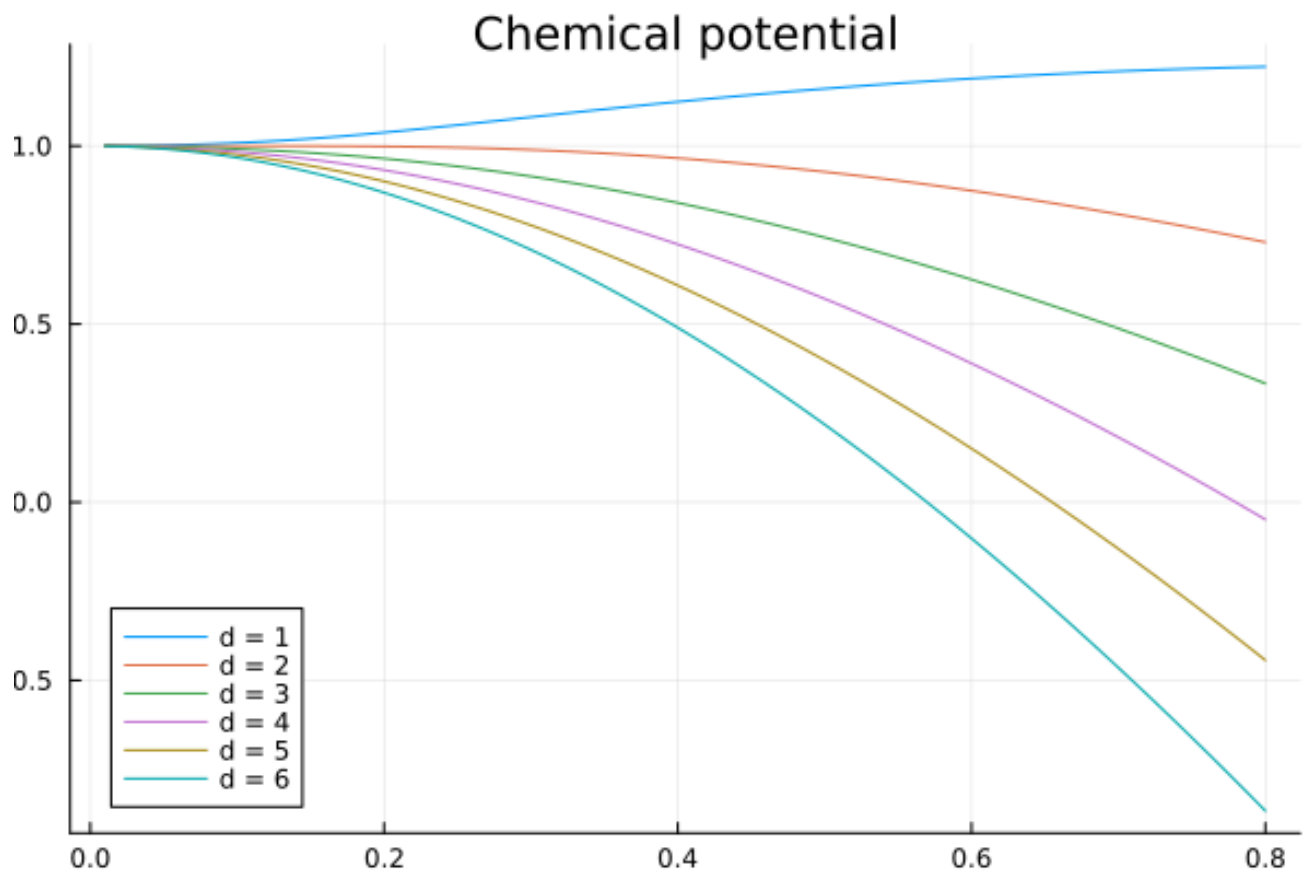
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Your Institution

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What controls the gas

The sign of the slope flips with α

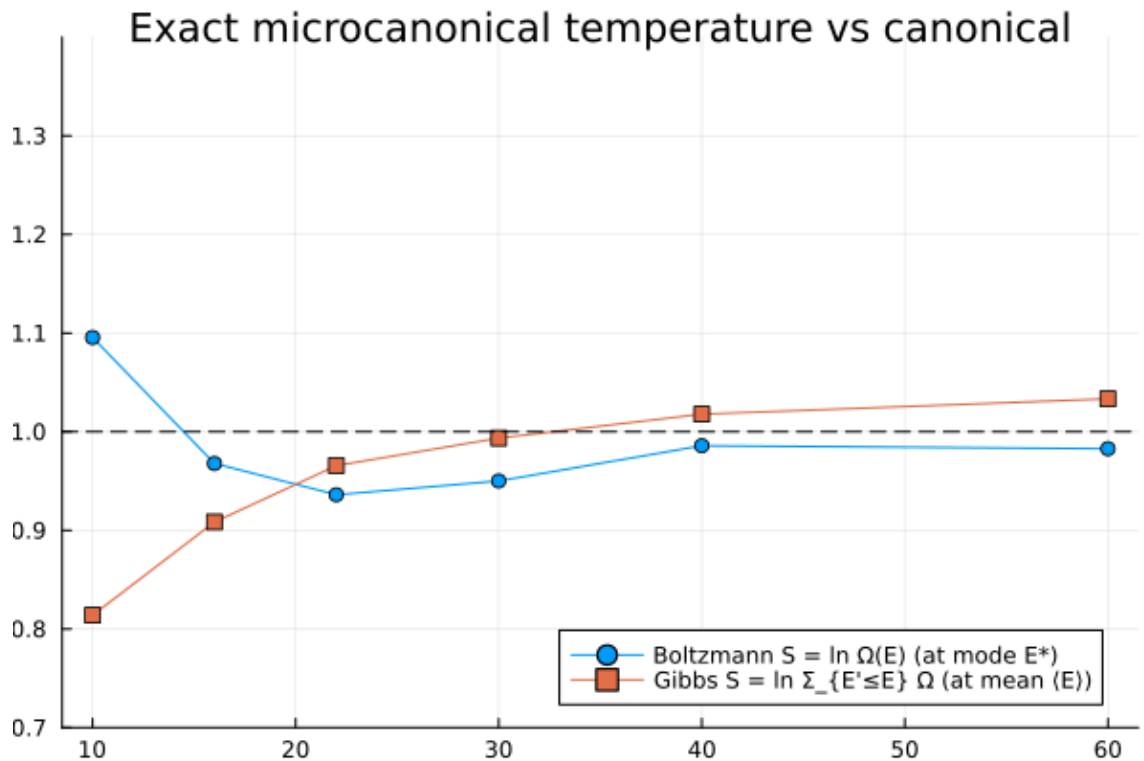


The sign of the slope flips with α

$d = 1$ *rises.* $d = 2$ *is flat.* $d \geq 3$ *falls.*

Temperature from state counts

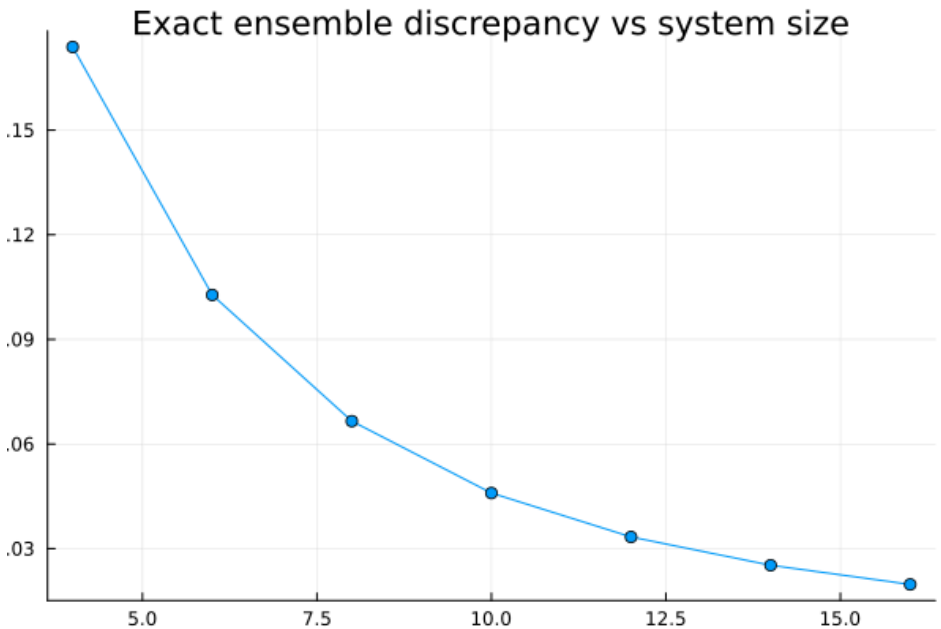
Boltzmann against Gibbs: the estimator sets the offset



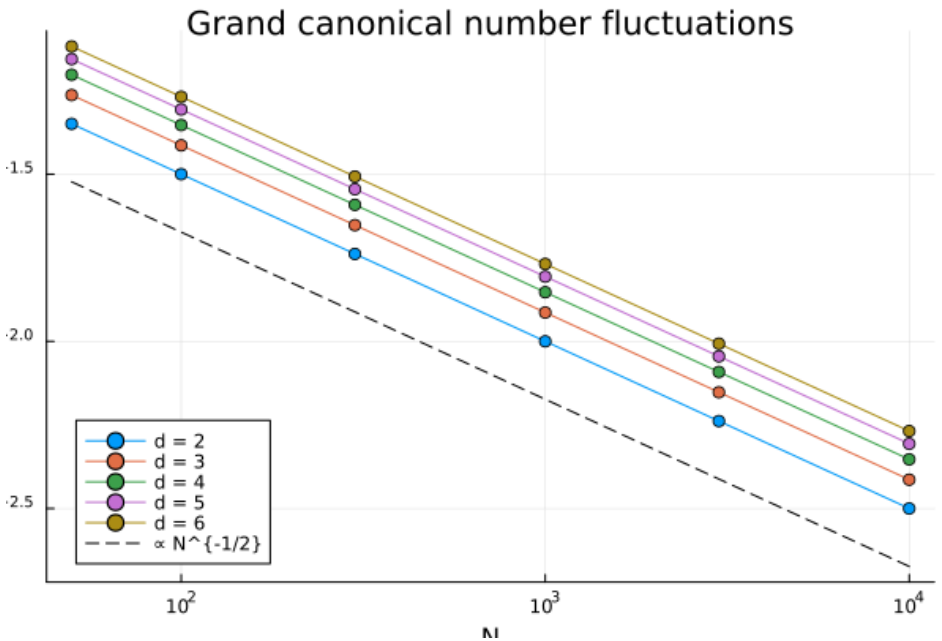
Beyond $L = 60$ *the two conventions bracket* T_0.

Do the ensembles agree?

Two of the three real checks



Exact GC against C, four-level box upward.



$\Delta N/N$ *at fixed* $T/\varepsilon_F = 0.1$.

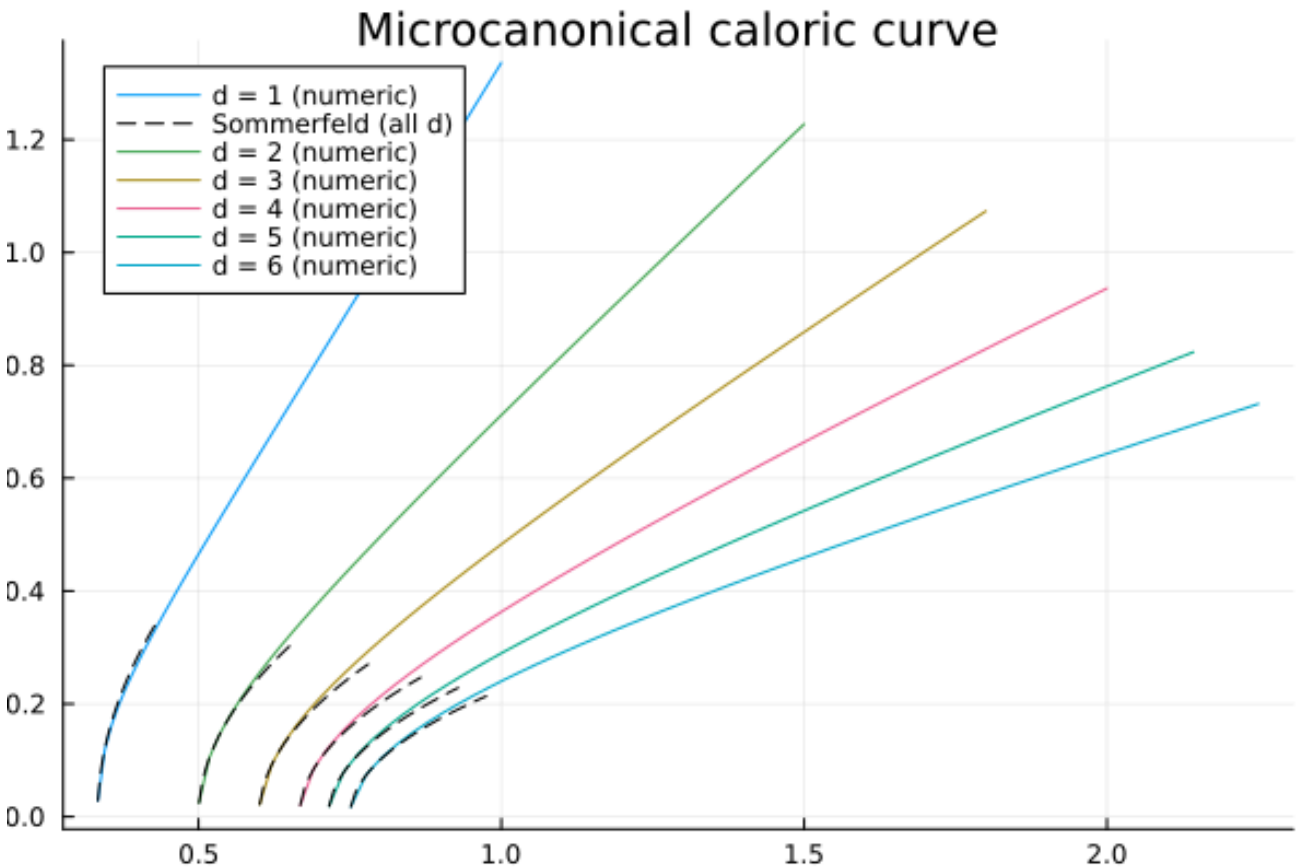
Questions?

Questions?

email@example.com

github.com/username

Backup: caloric curves



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Numeric inversion against the inverted Sommerfeld $U(T)$.